Proposal – Compositional Longitudinal Missing Data Imputation

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Problem

Design an imputation algorithm for missing data with 3 characteristics:

- **Compositional:** Sum-to-one row constraint
- **Longitudinal:** Multiple collected time points per subject
- **Sparsity:** Count data with high proportion of zeros

This combination of characteristics appears in many real-world datasets, e.g., microbiome data, Covid-19 variant data.



Literature Review

Existing imputation methods often focus on one or two of these characteristics but rarely all three simultaneously. For example:

- **General Compositional imputation** (Hron et al. [5], Saha et al. [7], Tsagris et al. [10]): introduce row-wise independent imputation methods for compositional data.
- **Longitudinal Compositional imputation for Microbiome data** (Choi et al. [2], Seki et al. [8, 9]): design compositional longitudinal imputation methods specifically for microbiome datasets, leveraging temporal correlations and phylogenetic information.

  $\rightarrow$ *This line of research is most relevant to our interested problem.*



Research Gap

Choi et al. [2], Seki et al. [8, 9] employed a common framework:

1. **Log-ratio (LR) transformation**: relax compositional constraints.
2. **Time-series multivariate imputation**: employ GAN [2], diffusion [8, 9] architectures to impute missing values on the transformed space.
3. **Inverse transformation**: transform the imputed values back to the original composition.

Research Gap: These studies assume row-wise missingness, which underestimates 2 painpoints – as LR encodes relative information:

1. It requires fully observed rows.
2. Its inverse transformation does not preserve the observed values in partially observed rows.



Explorations

- **Multiple Imputation (MI) framework**: The first gap can be mitigated by employing the MI framework, which assumes a temporarily imputed dataset is available for each iteration, allowing the LR transformation to be applied.
- **Addressing the second gap**: Two potential approaches are:
  1. Explore alternative transformations that retain the observed values during reconstruction.
  2. Change the imputation strategy to directly predict the composition, bypassing the need for inverse transformation.

Exploration – Alternative transformations

1. **Component-wise square-root** [3]: project composition onto the unit sphere:

$$z = (\sqrt{x_1}, \sqrt{x_2}, \ldots, \sqrt{x_D})$$

2. **Radial transformation** [6]: project composition onto the unit sphere after applying a radial transformation:

$$z = \frac{x}{\|x\|}$$

3. **Triangular Decoupling** [1]: transform the composition into a set of independent components using a triangular matrix decomposition:

$$x_1 = z_1, x_2 = (1 - z_1)z_2, \ldots, x_D = z_D \prod_{i=1}^{D-1} (1 - z_i)$$

Transformations 1 & 2: preserve zero values, relax compositional but introduce additional constraints.



Exploration – Directly Predicting Compositions

- **Idea**: Use the transformed data to directly predict the compositions.
- **Supporting theory**: Garcia et al. [4] – Models trained directly on raw compositions (response variable) outperform their CLR-transformed counterparts.
- **Expected flow**:
  1. Initialize imputed data X_0 using a general compositional imputation method [5, 7, 10] and set $t = 0$.
  2. Apply LR transformation: $X_t \mapsto Z_t$.
  3. Fit model f using Z_t to predict X_{t+1}.
  4. Check stopping condition: compare X_{t+1} with X_t.
  5. If the condition is not met, set $t = t + 1$ and repeat the process from step 2. Otherwise, terminate the iteration.



Proposed Network

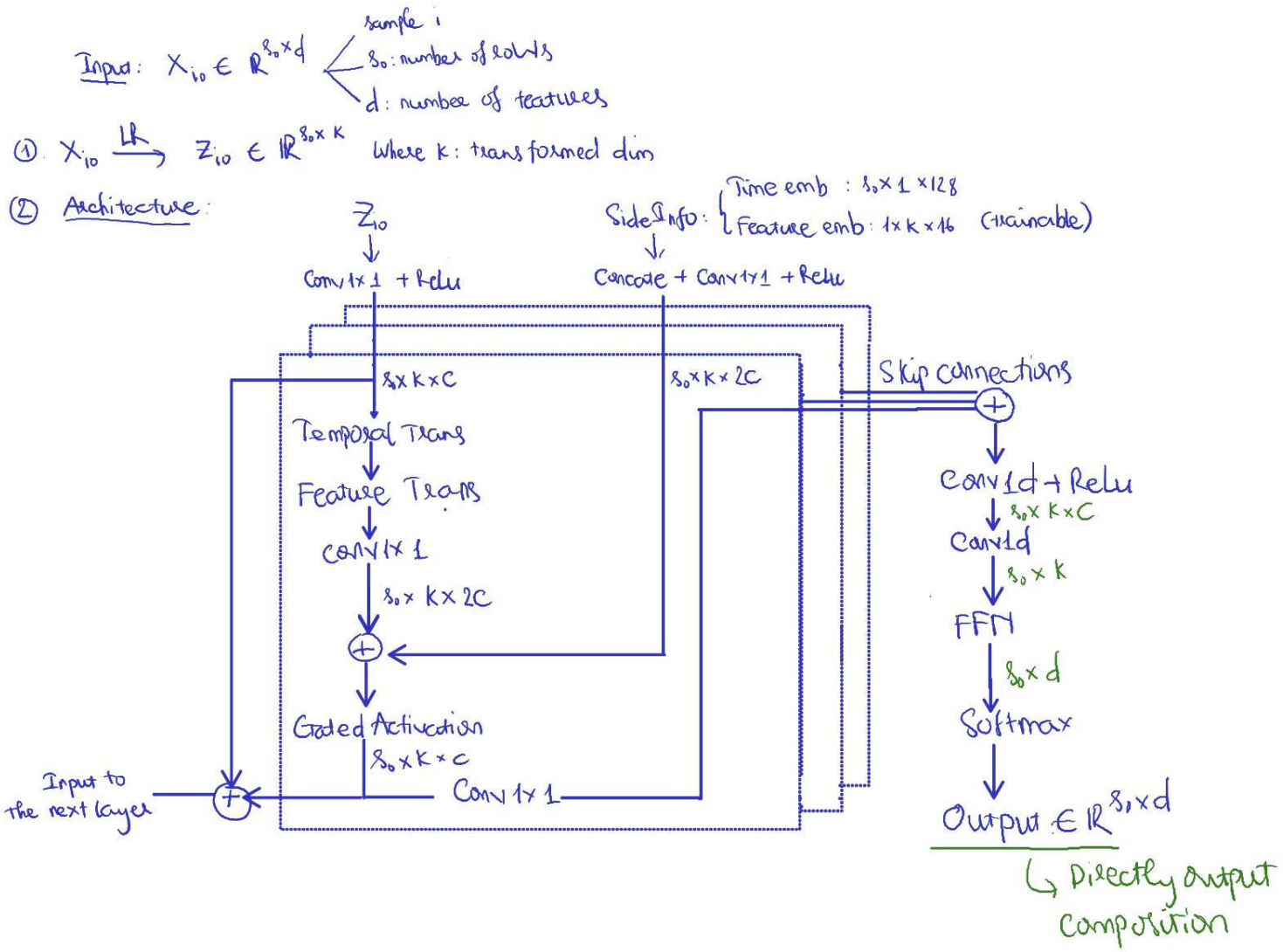


Figure 0.1: Proposed design for model f

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